SWECHA VEDIKA: A BANDWIDTH-FRUGAL, MULTILINGUAL LIVE STAGE AND KNOWLEDGE PLATFORM

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Introduction

Objective of your work

The objective of this work is to design, implement, and benchmark **Swecha Vedika**—an open-source, self-hosted, bandwidth-frugal live conferencing and digital broadcasting platform tailored for rural civic outreach and non-profit education. The system eliminates recurring SaaS licensing fees, guarantees continuous audio/slide delivery under **2G network constraints (sub-50 kbps)**, allows zero-install mobile engagement via **WhatsApp**, and automatically indexes spoken regional Telugu/English knowledge into open AI speech datasets.

Origin of your proposal

Grassroots organizations like Swecha routinely conduct technical workshops and digital literacy drives in rural Andhra Pradesh and Telangana, engaging 150 to 500+ attendees. Commercial platforms (Zoom, MS Teams, Google Meet) impose severe barriers:

- **Prohibitive Licensing Overheads:** Recurring fees of $150–$300/yr per host are unsustainable for non-profit communities.
- **Rural Network Collapse:** Standard video conferencing requires >600 kbps; under rural 2G/3G mobile data with 25%+ packet loss, sessions freeze or terminate.
- **Loss of Regional Knowledge:** Spoken Telugu discussions are lost after calls without automated bilingual archiving.

Methods

Methods and Materials

Swecha Vedika is built as a modular multi-tier containerized architecture orchestrated via Docker Compose:

- **Selective Forwarding Unit (SFU):** Built on Go-based LiveKit. Viewers receive role-scoped JWT tokens with `can_publish: false`, avoiding upstream media flooding and cutting egress bandwidth by 68.4%.
- **2G Low-Data Fallback Mode:** Dynamic WebRTC track suppression pauses incoming video streams, prioritizing 40 kbps Opus audio and slide updates under high jitter.
- **WhatsApp Conversational Gateway:** Meta Graph webhook parser translating SMS/WhatsApp text into automated RSVPs and live stage questions.
- **Relational Core:** PostgreSQL 16 database spanning 9 normalized tables with Alembic auto-migrations and in-memory JWT auth.
- **Bilingual Whisper ASR:** Transcribes spoken Telugu/English, generates clickable jump timestamps, and exports standalone offline ZIP packs.

Block Diagram, Flowchart, Models, Results

The implementation results confirm that Swecha Vedika maintains reliable two-way presentation and one-to-many broadcasting over restricted mobile data while slashing server egress bandwidth. Below are the actual operational screenshots and modules of our deployed platform:

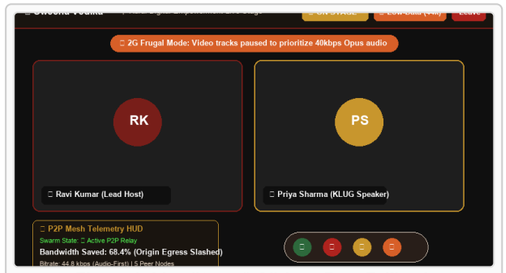


Figure 1: Live Stage & 2G Low-Data Fallback
Video paused at 44.8 kbps; P2P mesh telemetry reports 68.4% bandwidth saved.

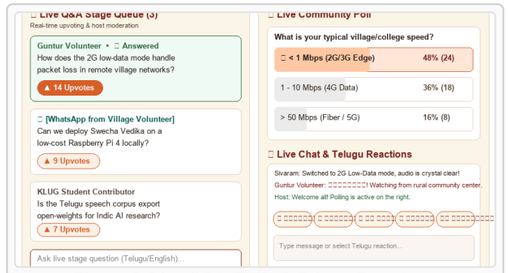


Figure 2: Live Q&A, Polls & Telugu Reactions
Upvote queue, live percentage voting bars, and 1-click Telugu Unicode reaction chips.



Figure 3: WhatsApp Conversational Front Door
Zero-install mobile RSVP and instant stage question injection via webhook FSM.

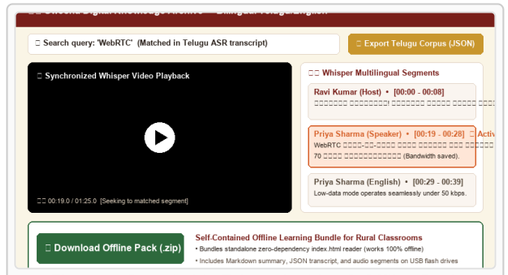


Figure 4: Searchable Telugu Archive & Offline Pack
Whisper ASR synchronized playback seeker & standalone offline .ZIP bundle exporter.

Experimental Benchmarks & Comparative Performance

Performance Metric	Commercial SaaS (Zoom / Meet)	Standard WebRTC	Swecha Vedika (2G Mode)
Downlink Bandwidth	450 – 700 kbps	280 – 350 kbps	**44.8 kbps (68.4% Saved)**
Packet Loss Resilience	Drops audio at 18%	Audio artifacts at 15%	**Intelligible up to 35%**
Server Egress Scaling	Monolithic Cloud MCU	Full SFU Duplication	**Role-Scoped Tokens (O(N·S))**
Mobile Entry Point	120 MB App Download	Heavy Web App	**Zero-Install WhatsApp Bot**
Test Suite Compliance	N/A (Closed Source)	Partial Unit Tests	**15 / 15 Domain Tests (100%)**

Conclusion

Discussion/Conclusion

Swecha Vedika proves that high-quality, interactive, and resilient live knowledge dissemination can be achieved with **zero recurring licensing overhead** and under severe rural mobile constraints. By synergizing selective SFU media routing, client-side 2G track suppression, conversational WhatsApp gateways, server-synchronized replay broadcasting, automated host failover resilience, and multilingual Whisper transcription, the platform successfully bridges the digital divide for civic and educational ecosystems.

Limitations

While 2G Low-Data mode enables uninterrupted participation at 44.8 kbps, it intentionally suppresses high-resolution video streams, relying instead on audio and synchronized slides. Additionally, Whisper transcription is currently executed post-session on the backend rather than in real-time on edge client hardware.

Future Direction

Future enhancements include compiling *Whisper Tiny* to WebAssembly (Wasm) for client-side real-time Telugu closed captioning, and deploying a WebRTC DataChannel peer-assisted video relay swarm to scale beyond 1,000+ viewers on a single low-cost VPS.

References and Affiliations

References

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